

Layered Architecture in Computer Networks

Layered architecture in computer networks refers to dividing a network's functioning into different layers, each responsible for a certain communication component. The major goal of this layered architecture is to separate the complex network communication process into manageable, smaller activities that can be better developed.

Therefore layered architecture provides interactions between the sub-systems. If any type of modification is done in one layer it does not affect the next layer.

Features of Layered Architecture

- Use of Layered architecture in computer network provides with the feature of modularity and distinct interfaces.
- Layered architecture ensures independence between layers, by offering services to higher layers from the lower layers and without specifying how these services are implemented.
- Layered architecture segments as larger and unmanageable design into small sub tasks.
- In layer architecture every network has different number of functions, layers and content.
- In layered architecture, the physical route provides with communication which is available under the layer 1.
- In layered architecture, the implementation done by one layer can be modified by another layer.

Elements of Layered Architecture

There are three different types of elements of a layered architecture. They are described below:

- **Service:** Service is defined as a set of functions and tasks being provided by a lower layer to a higher layer. Each layer performs a different type of task. Therefore, actions provided by each layer are different.
- **Protocol:** Protocol is defined as a set rule used by the layer for exchanging and transmission of data with its peer entities. These rules can consists details regarding a type of content and their order passed from one layer to another.
- **Interface:** Interface is defined as a channel that allows to transmit the messages from one layer to the another.

Significance of Layered Architecture

- **Divide and Conquer Approach:** Layered architecture supports divide and conquer approach. The unmanageable and complex task is further divided into smaller sub tasks. Each sub task is then carried out by the different layer.

Therefore, using this approach reduces the complexity of the problem or design process.

- **Easy to Modify:** The layers are independent of each other in layered architecture. If any sudden change occurs in the implementation of one layer, it can be changed. This change does not affect the working of other layers involved in the task. Therefore, layered architectures are required to perform any sudden update or change.
- **Modularity:** Layered architecture is more modular as compared to other architecture models in [computer network](#). Modularity provides with more independence between the layers and are easier to understand.
- **Easy to Test:** Each layer in layered architecture performs a different and dedicated task. Therefore, each layer can be analyzed and tested individually. It helps to analyze the problem and solve them more efficiently as compared to solving all the problems at a time.
- **Scalability:** As networks grow in size and complexity, additional layers or protocols may be added to meet new requirements while maintaining existing functionality.
- **Security:** The layered technique enables security measures to be implemented to varying degrees, protecting the community from a variety of threats.
- **Efficiency:** Each layer focuses on a certain aspect of verbal exchange, optimizing aid allocation and performance.

Evolution of Layered Architecture

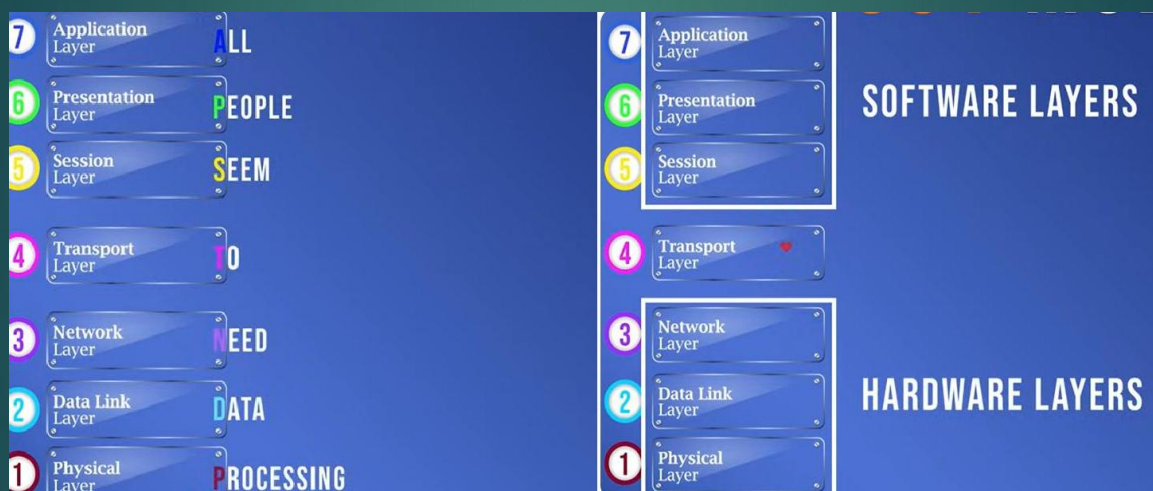
In computer networks, layered architecture is majorly used for communication. The two network models that makes use of layered architecture are:

- *OSI Model*

- **TCP/IP Model**



1. **OSI Model** The **OSI (Open Systems Interconnection)** Model is a set of rules that explains how different computer systems communicate over a network. OSI Model was developed by the International Organization for Standardization (ISO). The OSI Model consists of 7 layers and each layer has specific functions and responsibilities. When we transfer information from one device to another, it travels through 7 layers of OSI model. First data travels down through 7 layers from the sender's end and then climbs back 7 layers on the receiver's end.



- **Application Layer:** The Application Layer of OSI (Open System Interconnection) model, is the top layer in this model and takes care of network communication. The application layer provides the functionality to send and receive data from

users. It acts as the interface between the user and the application. The application provides services like file transmission, mail service, and many more.

1. Data Representation

It ensures that data is in a readable format for both the sender and receiver, including data translation, compression, and encryption.



2. Network Service Access

It provides applications with access to network services, enabling communication over a network.



3. Application Protocols

It supports protocols like HTTP, FTP, SMTP, and DNS that enable communication between applications.



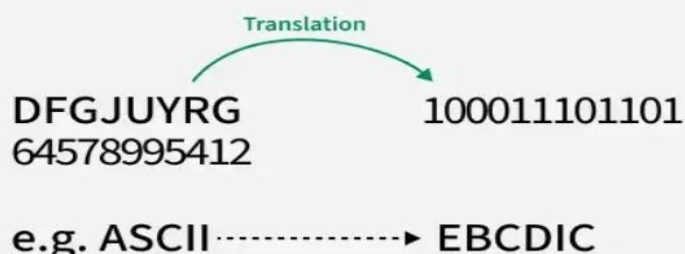
Application Layer



Presentation Layer is the 6th layer in the Open System Interconnection (OSI) model. This layer is also known as **Translation layer**, as this layer serves as a data translator for the network. The data which this layer receives from the Application Layer is extracted and manipulated here as per the required format to transmit over the network. The main responsibility of this layer is to provide or define the data format and encryption. The presentation layer is also called as **Syntax layer** since it is responsible for maintaining the proper syntax of the data which it either receives or transmits to other layer(s).

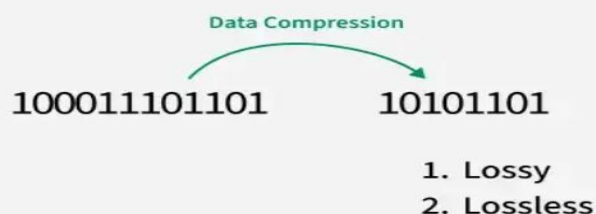
1. Data Translation

It converts data from the application layer into a network-suitable format and translates it back for the receiving application.(e.g., ASCII to EBCDIC)



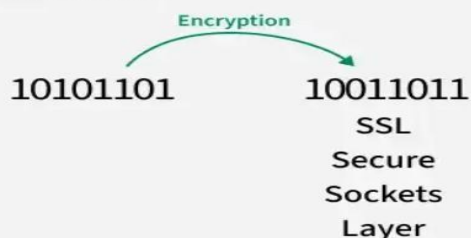
2. Data Compression

It compresses data to optimize bandwidth and speed, then decompresses it at the receiver's end. Examples: JPEG for images, MPEG for videos, and ZIP for files.



3. Data Encryption/Decryption

It secures communication by encrypting data before transmission and decrypting it at the destination. Examples of encryption protocols include SSL and TLS.



4. Syntax and Semantics Management

The presentation layer ensures that the syntax (data structure) and semantics (meaning of data) are properly managed.

The Session Layer is the 5th layer in the Open System Interconnection (OSI) model which plays an important role in controlling the dialogues (connections) between computers. This layer is responsible for setting up, coordinating, and terminating conversations, exchanges, and dialogues between the applications

at each end. It establishes, manages, and terminates the connections between the local and remote applications.

1. Session Establishment

It establishes connection-oriented or connectionless sessions between communicating parties for reliable data transfer (e.g., files, remote login).



Session of Communication

2. Communication Synchronization

It adds synchronization checkpoints in long data streams to ensure error recovery and proper data transmission.

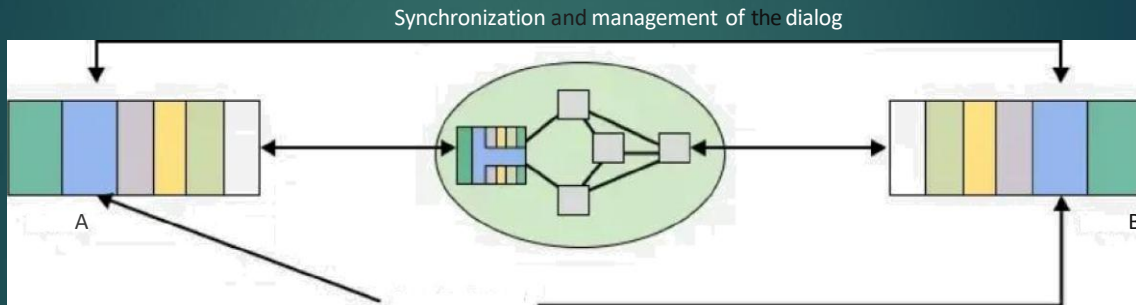


3. Activity Management

It splits message streams into logical units called "activities," allowing differentiation of tasks performed during a session.

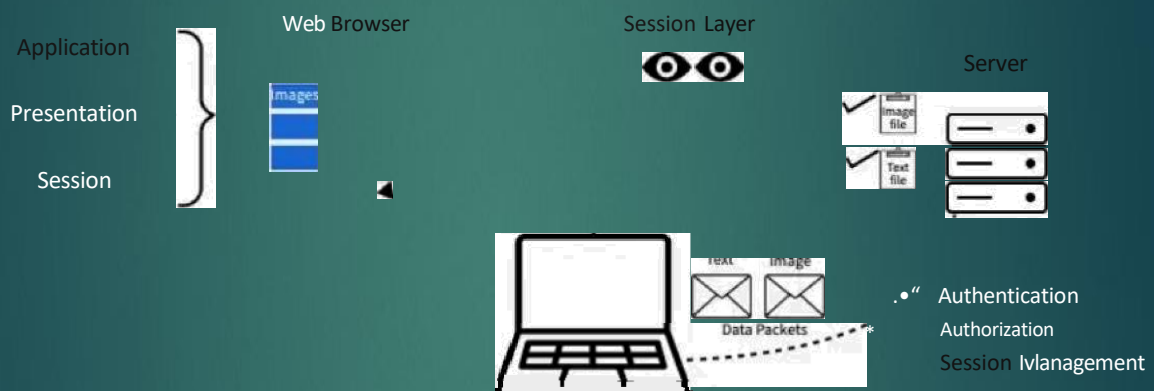
4. Dialog Management

It manages session dialogs, including initiating, maintaining, and ending communication.



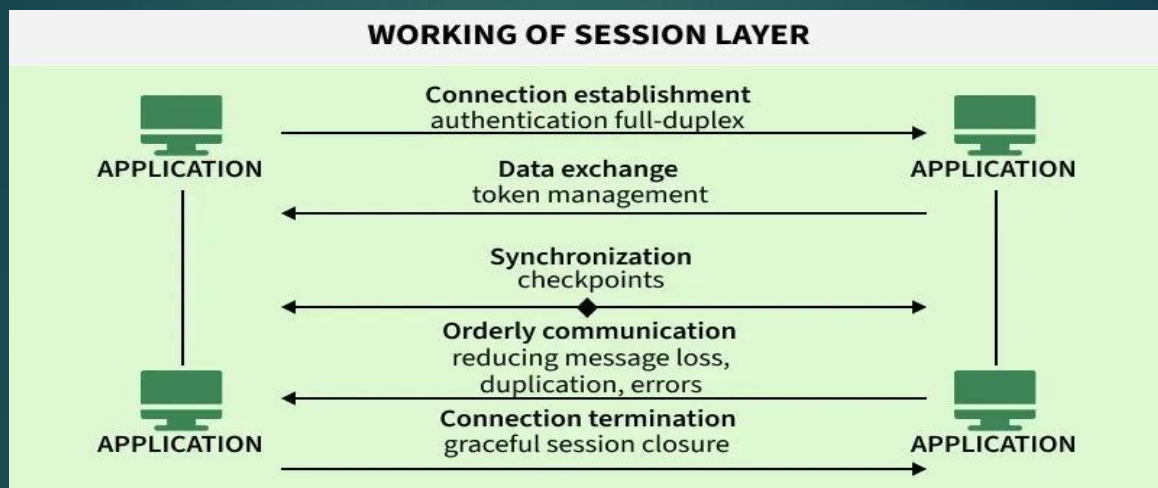
5. Data Transfer

It handles data exchange between users in; ensuring authentication and authorization.



6. Resynchronization

It restores session states to a previously defined synchronization point.

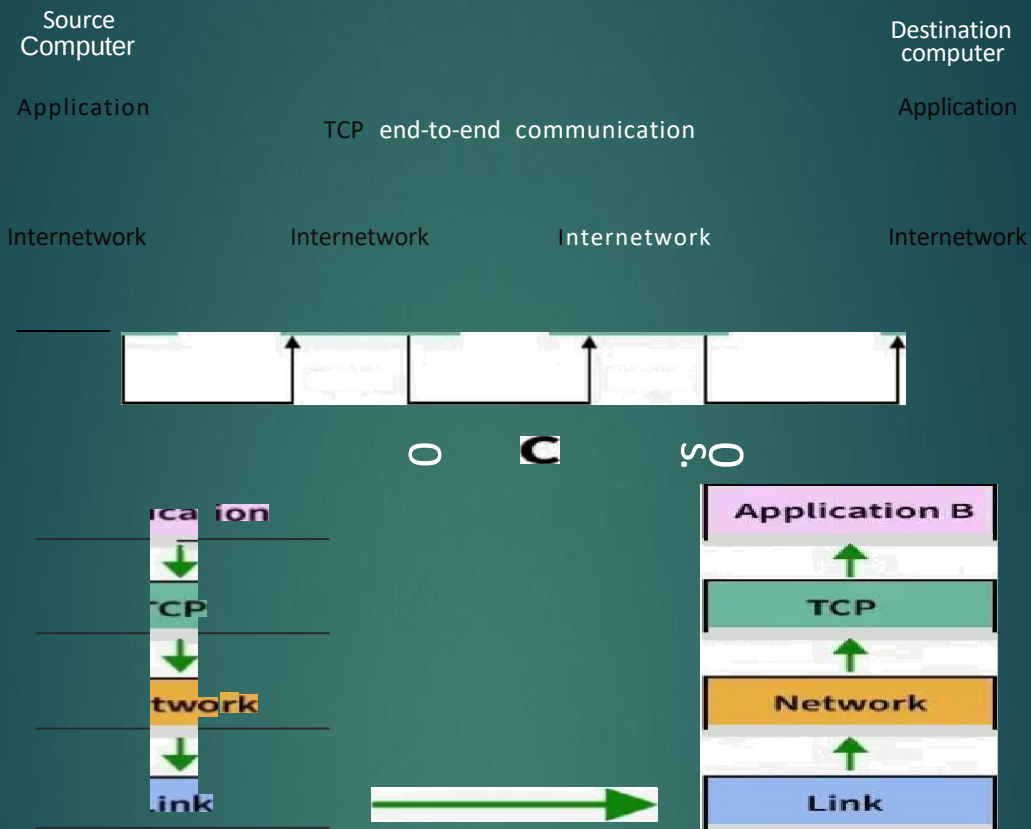


The transport layer, or layer 4 of the OSI model, controls network traffic between hosts and end systems to guarantee full data flows.

It is positioned between the network and session layers in the OSI paradigm. The data packets must be taken and sent to the appropriate machine by the network layer. After that, the transport layer receives the packets, sorts them, and looks for faults.

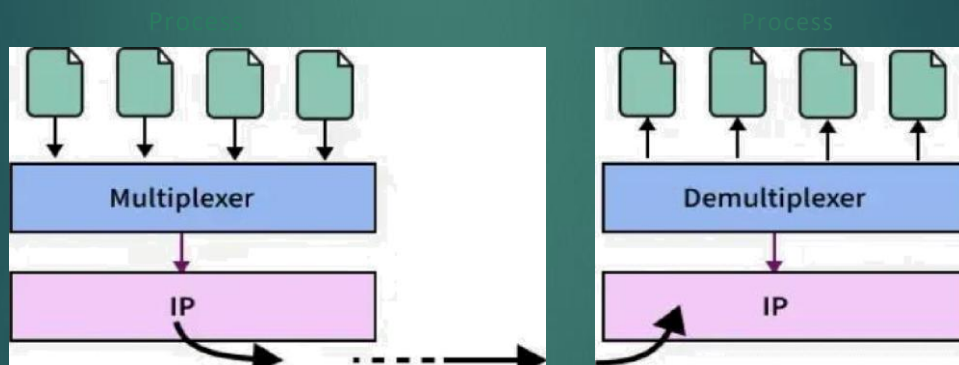
Subsequently, it directs them to the session layer of the appropriate computer program. Now, the properly structured packets are used by the session layer to hold the data for the application.

1. End-to-End communication



3. Multiplexing and Demultiplexing

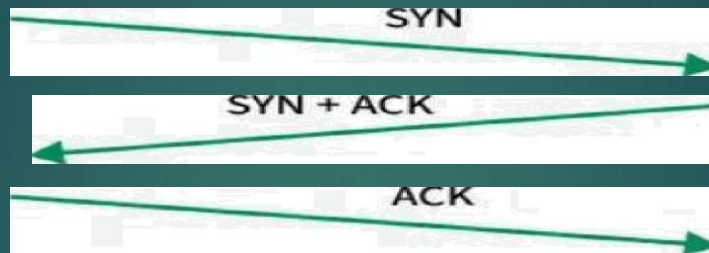
It manages multiple data streams on the same network connection by assigning unique ports.



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4. Connection Establishment* *



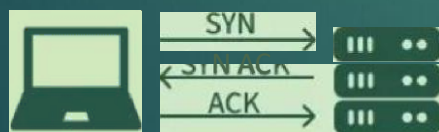
TCP 3 Way Hand shake

6. Reliable Data Delivery

It ensures all data is delivered correctly, in sequence, and without duplication (in protocols like TCP).

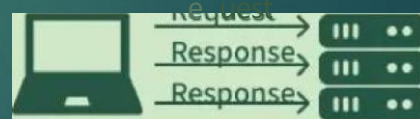
Transport Layer Protocols

TCP VS UDP



client

Server



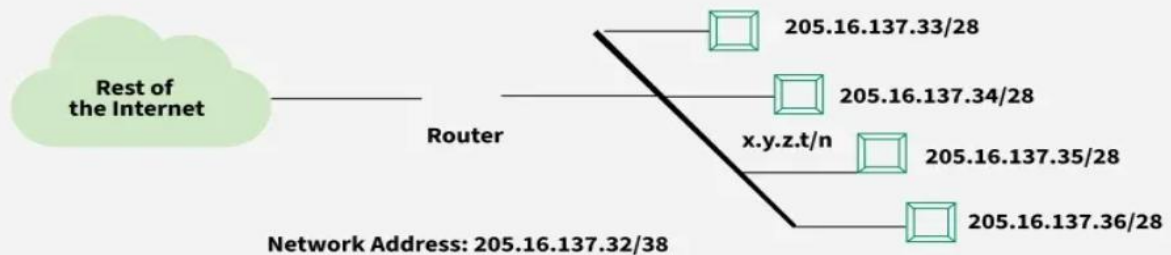
client

Server

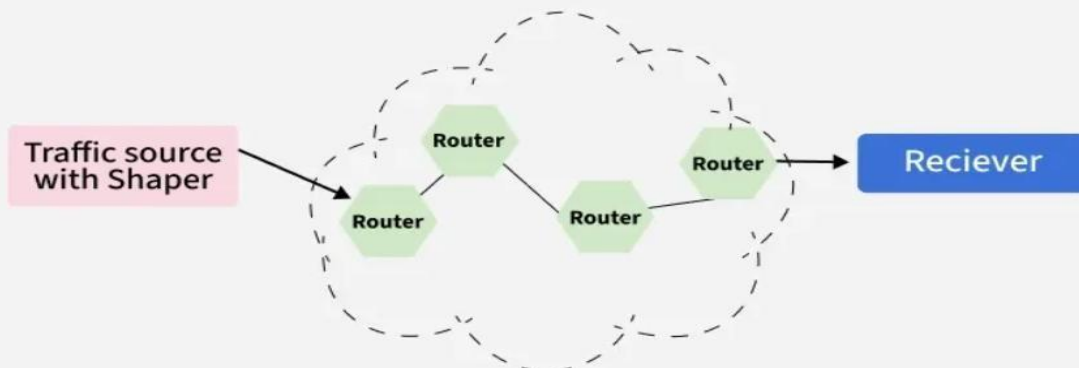
The Network Layer is the 5th Layer from the top and the 3rd layer from the Bottom of the [OSI Model](#). It is one of the most important layers which plays a key role in data transmission. The main job of this layer is to maintain the quality of the data and pass and transmit it from its source to its destination. It also handles [routing](#), which means that it chooses the best path to transmit the data from the source to its destination, not just transmitting the packet. There are several important protocols that work in this layer:

1. Assigning Logical Address

It assigns unique IP addresses to devices for identification across networks.



2. Packetizing



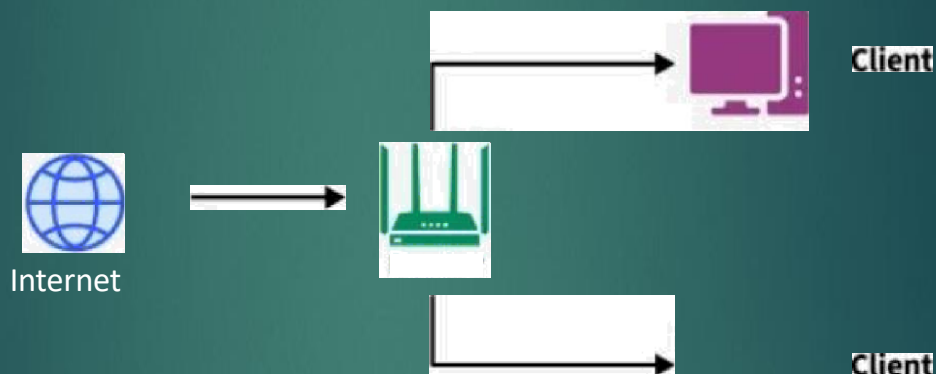
3. Host-to-host delivery

It ensures that data is reliably transmitted between two hosts across a network

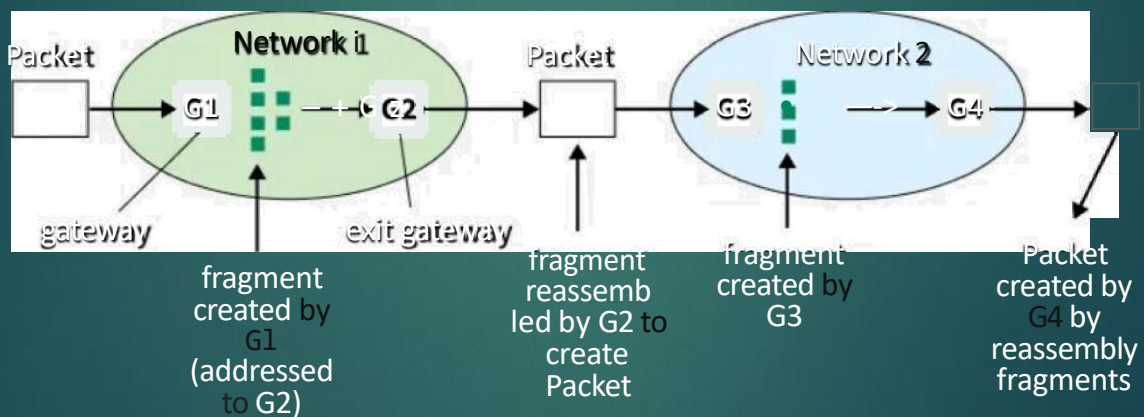


4. Forwarding

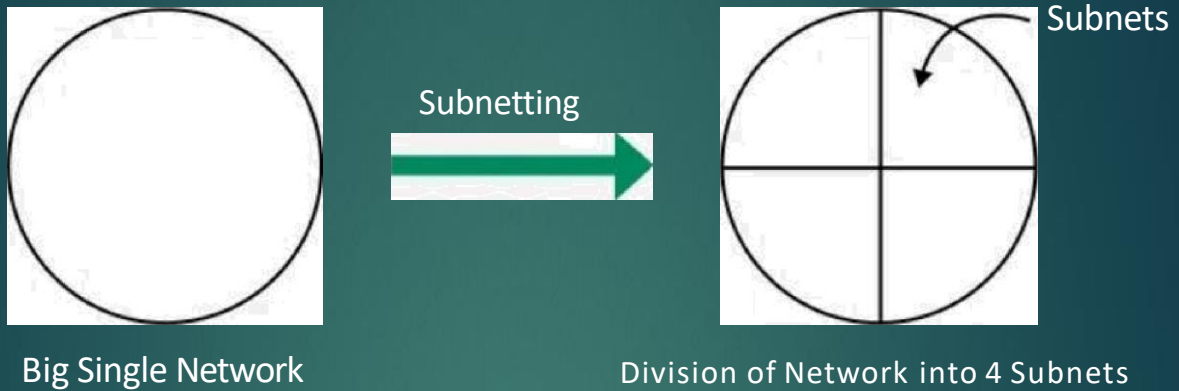
Forwarding moves packets through a router to the correct outgoing interface based on their destination.



5. Fragmentation and Reassembly of packets

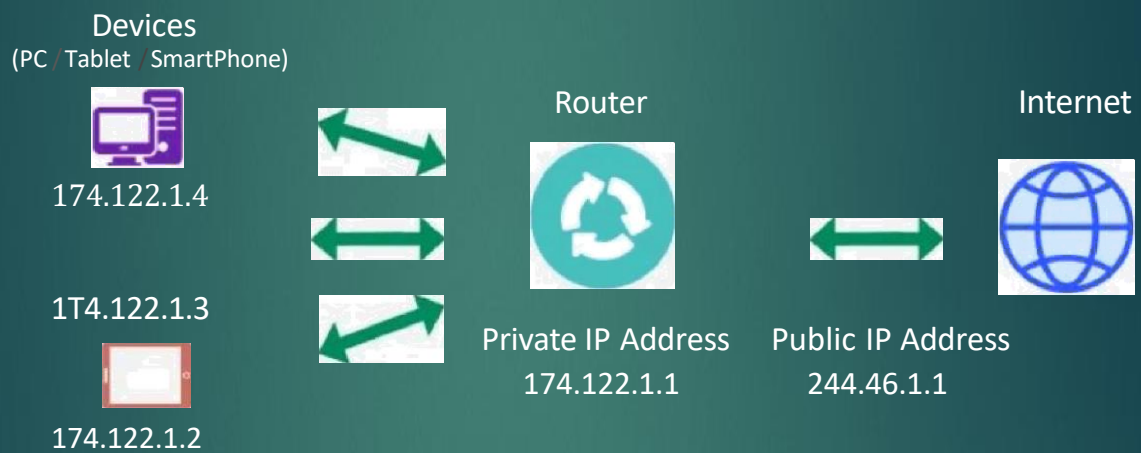


6. Logical Subnetting



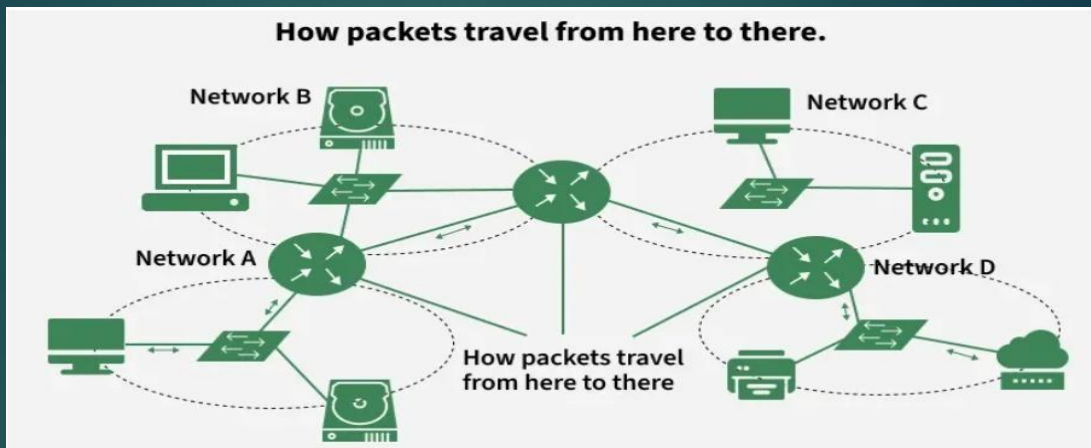
7. Network Address Translation

NAT is a method of mapping private IP addresses within a local network to a single public IP address (and vice versa)



8. Routing





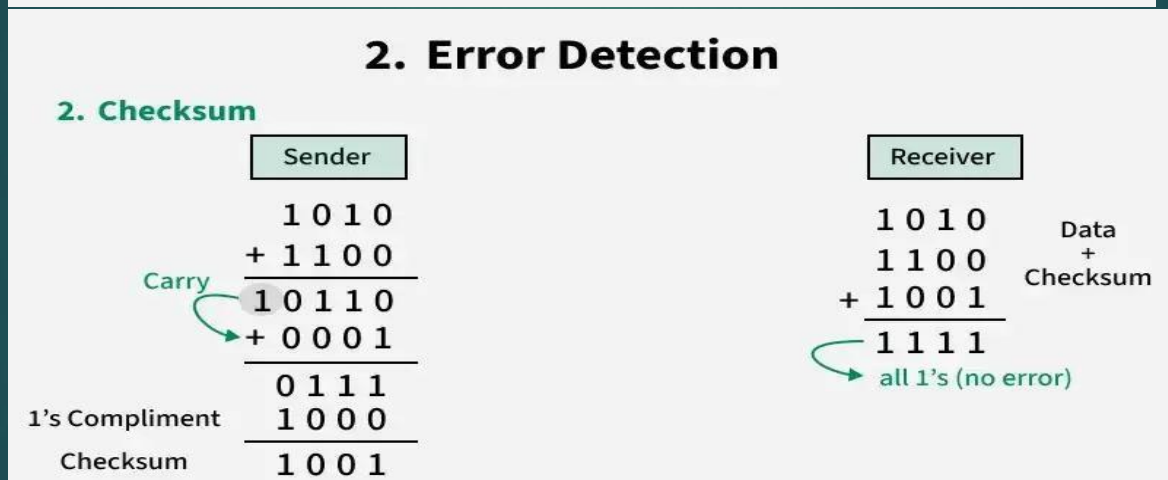
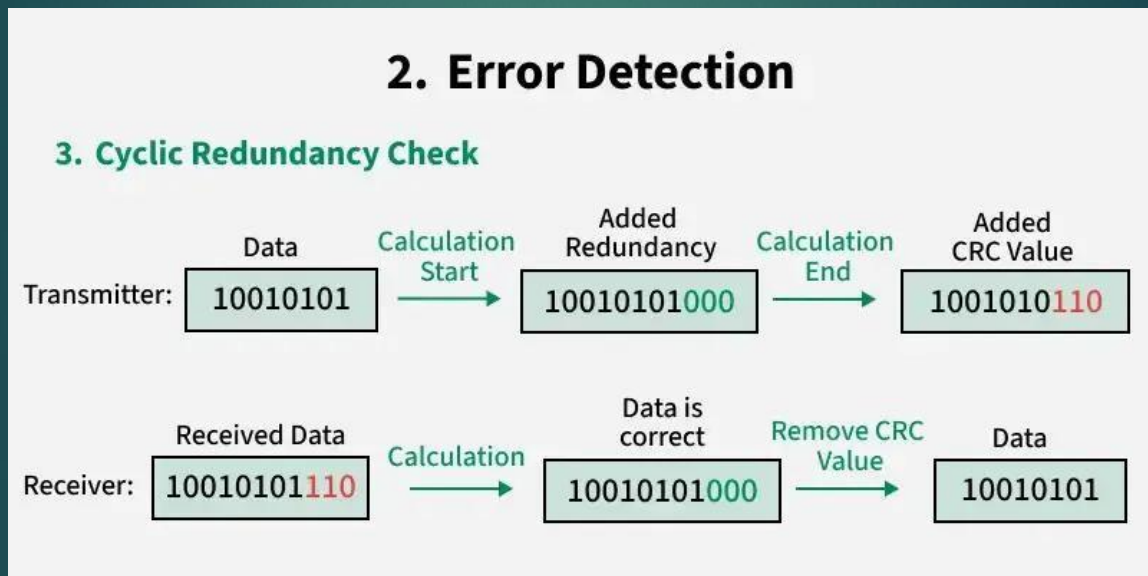
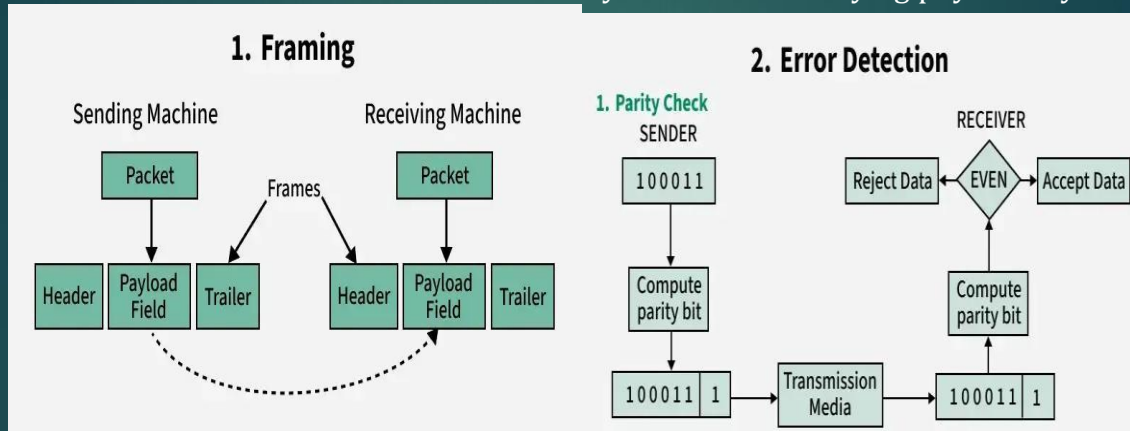
Every device gets a unique address (IP address) to identify it on the network.

- Data is packaged into small packets, with labels showing where it's coming from and where it's going.
- Routers figure out the best path to send the packets to their destination.
- Packets travel step by step through different routers until they reach the right device.
- If a packet is too big, it gets broken into smaller pieces to fit through the network.
- At the destination, the pieces are put back together into the original data.
- If something goes wrong, like the destination can't be reached, an error message is sent back.

Data link layer is the second layer from the bottom in the [OSI](#) (Open System Interconnection) network architecture model.

- Responsible for the node-to-node delivery of data within the same local network.
- Major role is to ensure error-free transmission of information.
- Also responsible for encoding, decoding, and organizing the outgoing and incoming data.
- Considered as the most complex layer of the OSI model as it hides all the underlying complexities of the hardware from the other above layers.
- **Sub-Layers of The Data Link Layer**
- The data link layer is further divided into two sub-layers, which are as follows:
- **Logical Link Control (LLC)**
- This sublayer of the data link layer deals with multiplexing, the flow of data among applications and other services, and LLC is responsible for providing error messages and acknowledgments as well.
- **Media Access Control (MAC)**

- MAC sublayer manages the device's interaction, responsible for addressing frames, and also controls physical media access. The data link layer receives the information in the form of packets from the Network layer, it divides packets into frames and sends those frames bit-by-bit to the underlying physical layer



3. Error Correction

- **Automatic-repeat request (ARQ)**

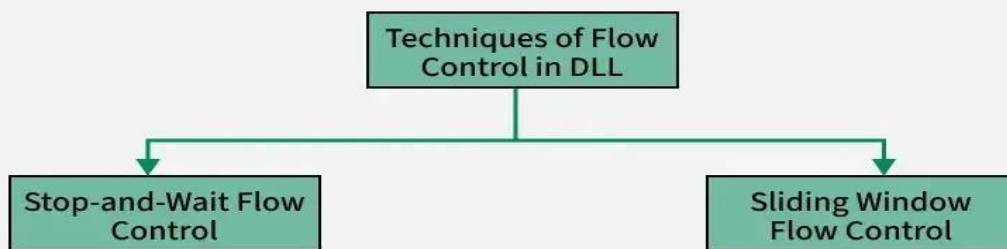
ARQ is a feedback-based method where the receiver requests the sender to retransmit the data if errors are detected.

- **Forward error correction (FEC):**

FEC is a method where the sender includes additional redundant data, allowing the receiver to detect and correct errors without retransmission.

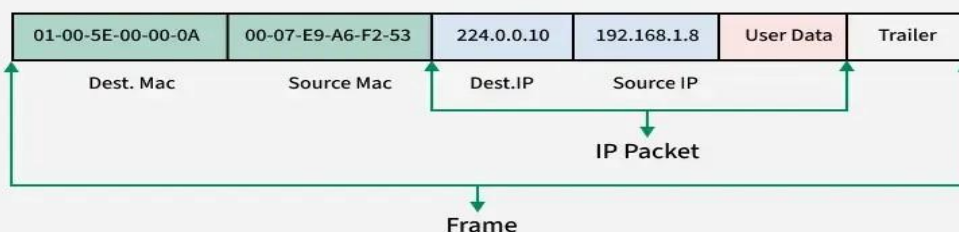
4. Flow Control

The Data Link Layer ensures flow control by synchronizing the sender's and receiver's speeds to prevent buffer overflow and frame loss.



5. Addressing

The Data Link Layer adds source and destination MAC addresses in the frame header for node-to-node delivery.

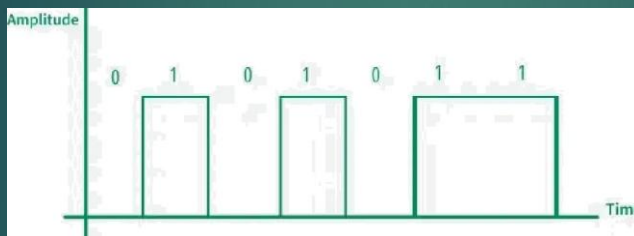


Physical Layer is the bottom-most layer in the Open System Interconnection (OSI) Model, responsible for the physical and electrical transmission of data. It consists of various network components such as power plugs, connectors, receivers, cable types, etc. The physical layer sends data bits from one device(s) (like a computer) to another device(s).

1. Bit-by-Bit Transmission



2. Encoding and Decoding



3. Signal Transmission

It is responsible for converting data into signals (analog or digital) for transmission and decoding these signals at the receiver's end.



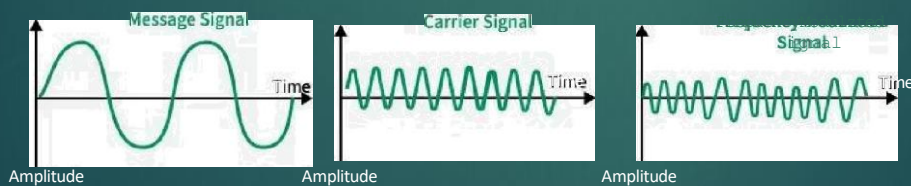
Digital Signals



Analog Signals

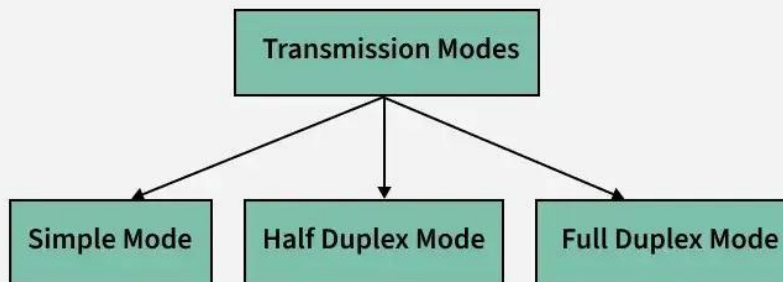
4. Modulation and Demodulation

Modulation is the process of modifying a carrier signal's properties (amplitude, frequency, or phase) for transmission over a communication medium.



5. Transmission Modes

Physical Layer determines the direction of data flow.



6. Data Rate Control

Physical Layer controls the speed of data transmission, ensuring the sender and receiver can handle the same data rate.

Let us suppose, **Person A** sends an e-mail to his friend **Person B**.

Step 1: Person A interacts with e-mail application like **Gmail, outlook**, etc. Writes his email to send. (This happens at **Application Layer**).

Step 2: At Presentation Layer, Mail application prepares for data transmission like encrypting data and formatting it for transmission.

Step 3: At Session Layer, there is a connection established between the sender and receiver on the internet.

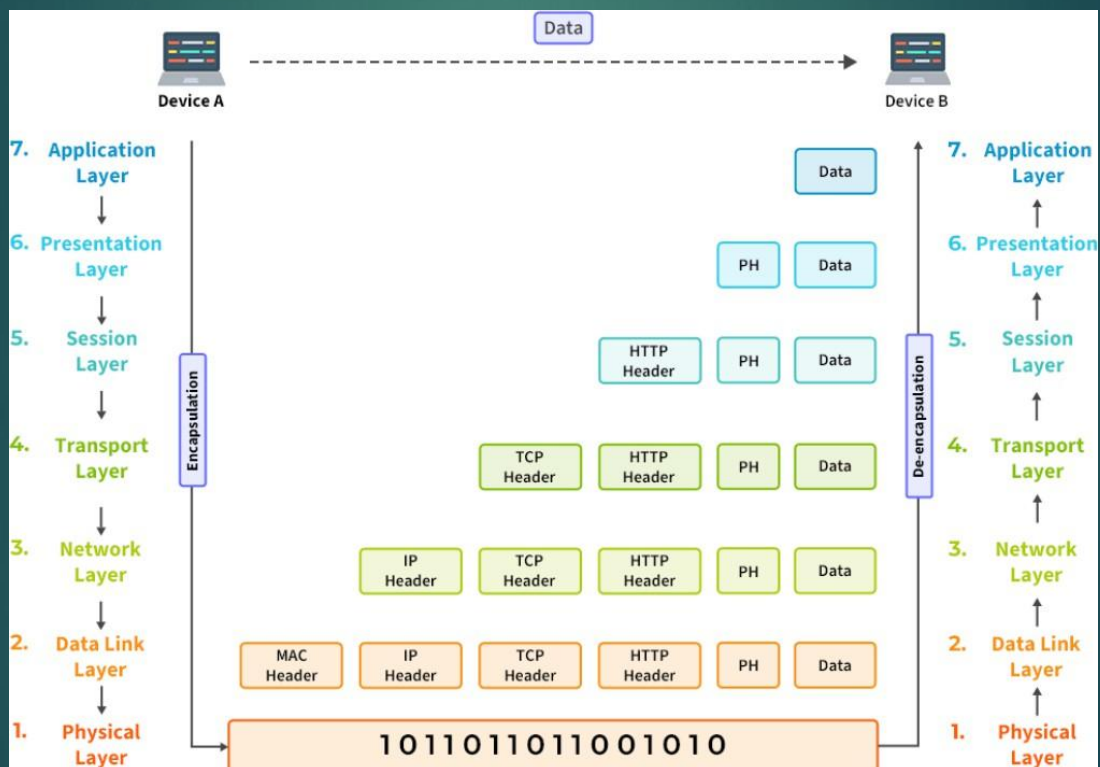
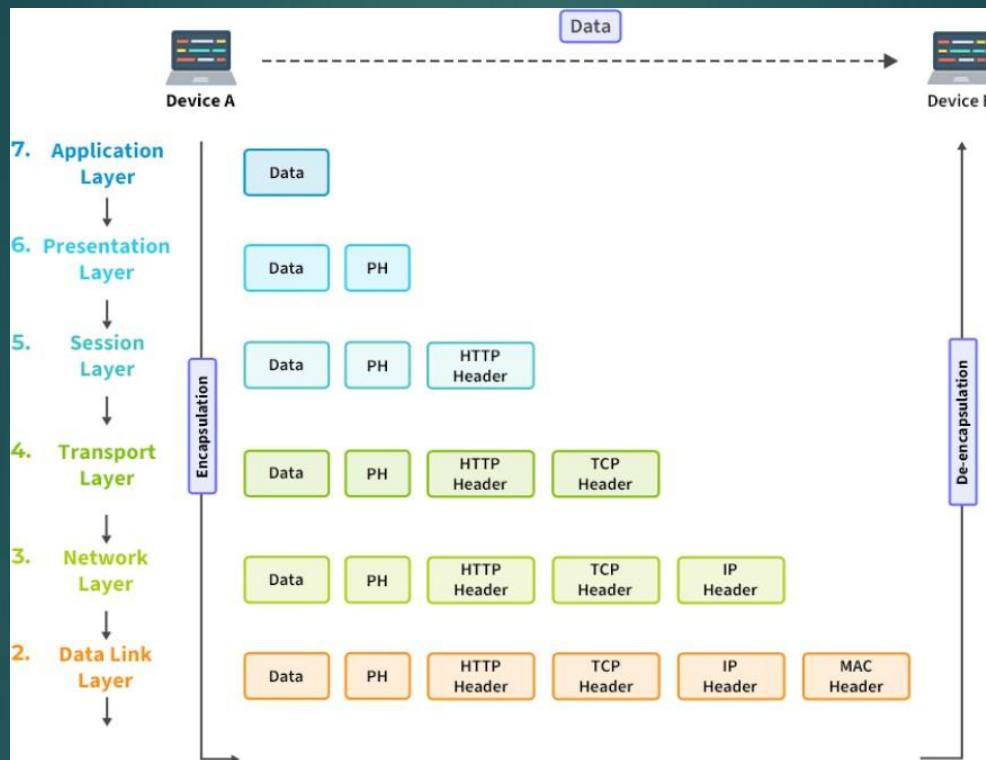
Step 4: At Transport Layer, Email data is broken into smaller segments. It adds sequence number and error-checking information to maintain the reliability of the information.

Step 5: At Network Layer, addressing of packets is done in order to find the best route for transfer.

Step 6: At Data Link Layer, data packets are encapsulated into frames, then MAC address is added for local devices and then it checks for error using error detection.

Step 7: At Physical Layer, Frames are transmitted in the form of electrical/ optical signals over a physical network medium like ethernet cable or WiFi.

After the email reaches the receiver i.e. **Person B**, the process will reverse and decrypt the e-mail content. At last, the email will be shown on **Person B** email client.



TCP/IP Model

1. Network Access Layer

It is a group of applications requiring network communications. This layer is responsible for generating the data and requesting connections. It acts on behalf of the sender and the Network Access layer on the behalf of the receiver. During this article, we will be talking on the behalf of the receiver.

The packet's network protocol type, in this case, TCP/IP, is identified by network access layer. Error prevention and "framing" are also provided by this layer. [Point-to-Point Protocol \(PPP\)](#) framing and Ethernet IEEE 802.2 framing are two examples of data-link layer protocols.

2. Internet Layer

This layer parallels the functions of OSI's Network layer. It defines the protocols which are responsible for the logical transmission of data over the entire network. The main protocols residing at this layer are as follows:

- **IP:** [IP](#) stands for Internet Protocol and it is responsible for delivering packets from the source host to the destination host by looking at the IP addresses in the packet headers. IP has 2 versions: IPv4 and IPv6. IPv4 is the one that most websites are using currently. But IPv6 is growing as the number of IPv4 addresses is limited in number when compared to the number of users.
- **ICMP:** [ICMP](#) stands for Internet Control Message Protocol. It is encapsulated within IP datagrams and is responsible for providing hosts with information about network problems.
- **ARP:** [ARP](#) stands for Address Resolution Protocol. Its job is to find the hardware address of a host from a known IP address. ARP has several types: Reverse ARP, Proxy ARP, Gratuitous ARP, and Inverse ARP.

3. Transport Layer

The TCP/IP transport layer protocols exchange data receipt acknowledgments and retransmit missing packets to ensure that packets arrive in order and without error. End-to-end communication is referred to as such. Transmission Control Protocol (TCP) and User Datagram Protocol are transport layer protocols at this level (UDP).

- **TCP:** Applications can interact with one another using [TCP](#) as though they were physically connected by a circuit. TCP transmits data in a way that resembles character-by-character transmission rather than separate packets. A starting point that establishes the connection, the whole transmission in byte order, and an ending point that closes the connection make up this transmission.
- **UDP:** The datagram delivery service is provided by [UDP](#) the other transport layer protocol. Connections between receiving and sending hosts are not verified by UDP. Applications that transport little amounts of data use UDP rather than TCP because it eliminates the processes of establishing and validating connections.

4. Application Layer

This layer is analogous to the transport layer of the OSI model. It is responsible for end-to-end communication and error-free delivery of data. It shields the upper-layer applications from the complexities of data. The three main protocols present in this layer are:

- **HTTP and HTTPS:** HTTP stands for Hypertext transfer protocol. It is used by the World Wide Web to manage communications between web browsers and servers. HTTPS stands for HTTP-Secure. It is a combination of HTTP with SSL(Secure Socket Layer). It is efficient in cases where the browser needs to fill out forms, sign in, authenticate, and carry out bank transactions.
- **SSH:** SSH stands for Secure Shell. It is a terminal emulations software similar to Telnet. The reason SSH is preferred is because of its ability to maintain the encrypted connection. It sets up a secure session over a TCP/IP connection.
- **NTP:** NTP stands for Network Time Protocol. It is used to synchronize the clocks on our computer to one standard time source. It is very useful in situations like bank transactions. Assume the following situation without the presence of NTP. Suppose you carry out a transaction, where your computer reads the time at 2:30 PM while the server records it at 2:28 PM. The server can crash very badly if it's out of sync.

Benefits of Layered Architecture

- Modularity
- Interoperability
- Flexibility
- Reusability
- Scalability
- Security

Challenges in Layered Architecture

- Performance Overhead
- Complexity in Implementation
- Resource Utilization
- Debugging and Troubleshooting
- Protocol Overhead

Real-World Applications and Examples

- Internet Browsing
- Email Communication
- Video Streaming

- Online Gaming
- Voice over IP (VoIP)
- Smart Home Devices
- Cloud Computing
- Blockchain Networks

Conclusion

Layered architecture in computer networks significantly simplifies the complex process involved in network communication by dividing them into smaller process, each with its own function and responsibility. This method improves modularity, interoperability, flexibility, and security, making network systems easier to design, manage, and expand. Despite problems like as performance overhead and implementation complexity, the layered model's benefits may be seen in real-world applications such as internet browsing, email communication, video streaming, and more, emphasising its importance in modern networking.

Networking Protocols

A Quick Overview

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HTTP
Hypertext Transfer Protocol

This is how web pages are requested and delivered on the internet.



HTTPS
Hypertext Transfer Protocol Secure

It's like HTTP, but with added security for safer data transfer.



FTP
File Transfer Protocol

It's used to send and receive files over the internet.



TCP
Transmission Control Protocol

It ensures that data sent over the internet arrives intact.



IP
Internet Protocol

It provides addresses for devices on the internet.



UDP
User Datagram Protocol

It's like TCP, but faster and less concerned with data integrity.



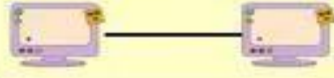
SMTP
Simple Mail Transfer Protocol

It's used for sending emails between servers.



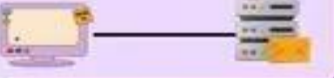
SSH
Secure Shell

It's a secure way to access and control remote computers.



IMAP
Internet Message Access Protocol

Allows users to access email messages stored on a remote mail server.



The OSI model organizes network communication into seven layers, and each layer uses specific protocols to perform its job. Here's a breakdown of the key protocols and their functions for each layer.

Layer 7: Application Layer

This layer is the user's interface to the network. It's responsible for providing network services directly to software applications.

- **HTTP (Hypertext Transfer Protocol):** Governs web Browse and communication between web servers and clients (browsers).
 - **HTTPS (Hypertext Transfer Protocol Secure):** The secure version of HTTP, it adds a layer of encryption for safe data transmission, like online banking.
 - **FTP (File Transfer Protocol):** Facilitates the transfer of files between a client and a server.
 - **SMTP (Simple Mail Transfer Protocol):** Used to send email from a client to a server.
 - **DNS (Domain Name System):** Translates human-readable domain names (e.g., google.com) into IP addresses that computers can understand.
-

Layer 6: Presentation Layer

The presentation layer handles data formatting, encryption, and compression to ensure data from one system can be read by another.

- **SSL/TLS (Secure Sockets Layer/Transport Layer Security):** While often associated with the Transport layer, these protocols start at the Presentation layer to handle data encryption and decryption, ensuring secure communication for protocols like HTTPS.
 - **JPEG, MPEG, GIF:** These are not traditional networking protocols but are examples of standards handled at this layer for data formatting (image and video encoding).
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Layer 5: Session Layer

This layer is responsible for establishing, managing, and terminating communication sessions between applications.

- **NetBIOS (Network Basic Input/Output System):** Allows applications on different computers to communicate over a local area network (LAN).
 - **RPC (Remote Procedure Call):** A protocol that allows a program on one computer to execute code on a different computer without the programmer needing to know the details of the remote network.
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Layer 4: Transport Layer

The transport layer manages end-to-end communication and ensures reliable data delivery. It segments data and provides flow and error control.

- **TCP (Transmission Control Protocol):** A **connection-oriented** protocol that ensures reliable, ordered, and error-checked data delivery. It's used for applications where data integrity is critical, like email and file transfers.
 - **UDP (User Datagram Protocol):** A **connectionless** protocol that is faster but less reliable than TCP. It doesn't guarantee delivery or order and is used for applications where speed is more important than accuracy, like video streaming and online gaming.
-

Layer 3: Network Layer

This layer handles **logical addressing and routing**. It determines the best path for data to travel across different networks.

- **IP (Internet Protocol):** The primary protocol of the internet, it provides logical addresses (**IP addresses**) to devices and is responsible for routing data packets from a source to a destination.
 - **ICMP (Internet Control Message Protocol):** Used by network devices to send error messages and operational information. The ping command relies on this protocol.
 - **ARP (Address Resolution Protocol):** Translates a logical IP address into a physical **MAC address** on a local network.
 - **Routing Protocols (e.g., RIP, OSPF, BGP):** These are used by routers to dynamically find the best paths for data to travel.
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Layer 2: Data Link Layer

The data link layer is responsible for node-to-node data transfer within a local network segment. It packages raw bits into **frames** and provides physical addressing.

- **Ethernet:** The most common protocol for wired local area networks (LANs). It defines the frame format and physical addressing (**MAC addresses**).
 - **PPP (Point-to-Point Protocol):** Used for establishing a direct connection between two networking nodes.
 - **Wi-Fi (IEEE 802.11):** A set of protocols that govern wireless local area networks.
 - **Switching:** While not a single protocol, the job of a network switch to forward frames based on MAC addresses is a key function of this layer.
-

Layer 1: Physical Layer

This is the lowest layer, dealing with the physical transmission of raw **bits** (1s and 0s) over a physical medium. It doesn't have a lot of complex protocols but rather specifies standards.

- **USB, Bluetooth, Ethernet Cable (e.g., Cat6):** These are physical standards and specifications that define the type of cabling, connectors, and electrical signals used to transmit data.
- **DSL, ISDN, T1:** These are protocols that define how digital data is transmitted over telephone lines.